Exploring Four Questions about Energy

→ Graphs made with R’s ggplot2

Key Questions

1. When will the world run out of oil?
2. Where does the world's oil come from?
3. Are wind and solar ready to power the electric grid?
4. What is Louisiana's role in the global energy economy?

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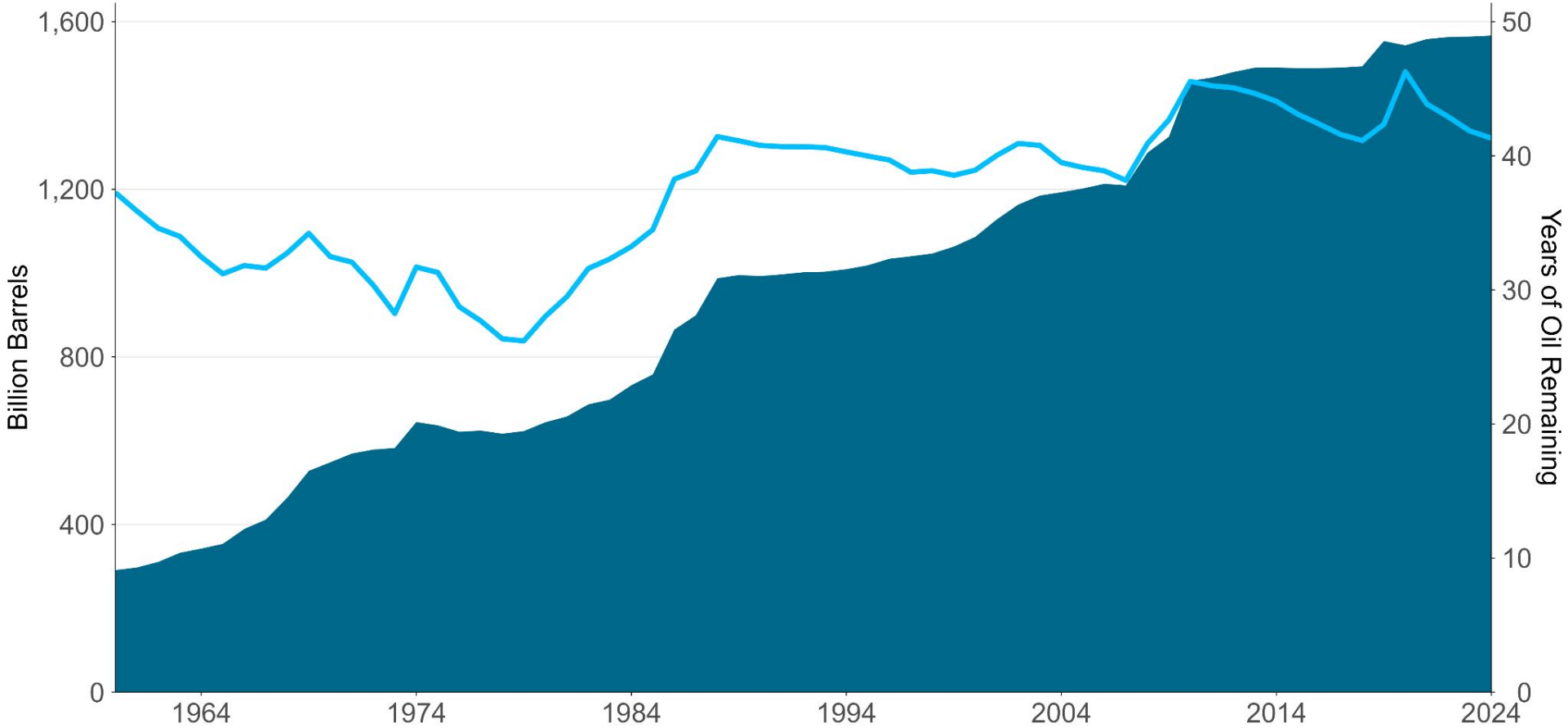
Global Proved Oil Reserves

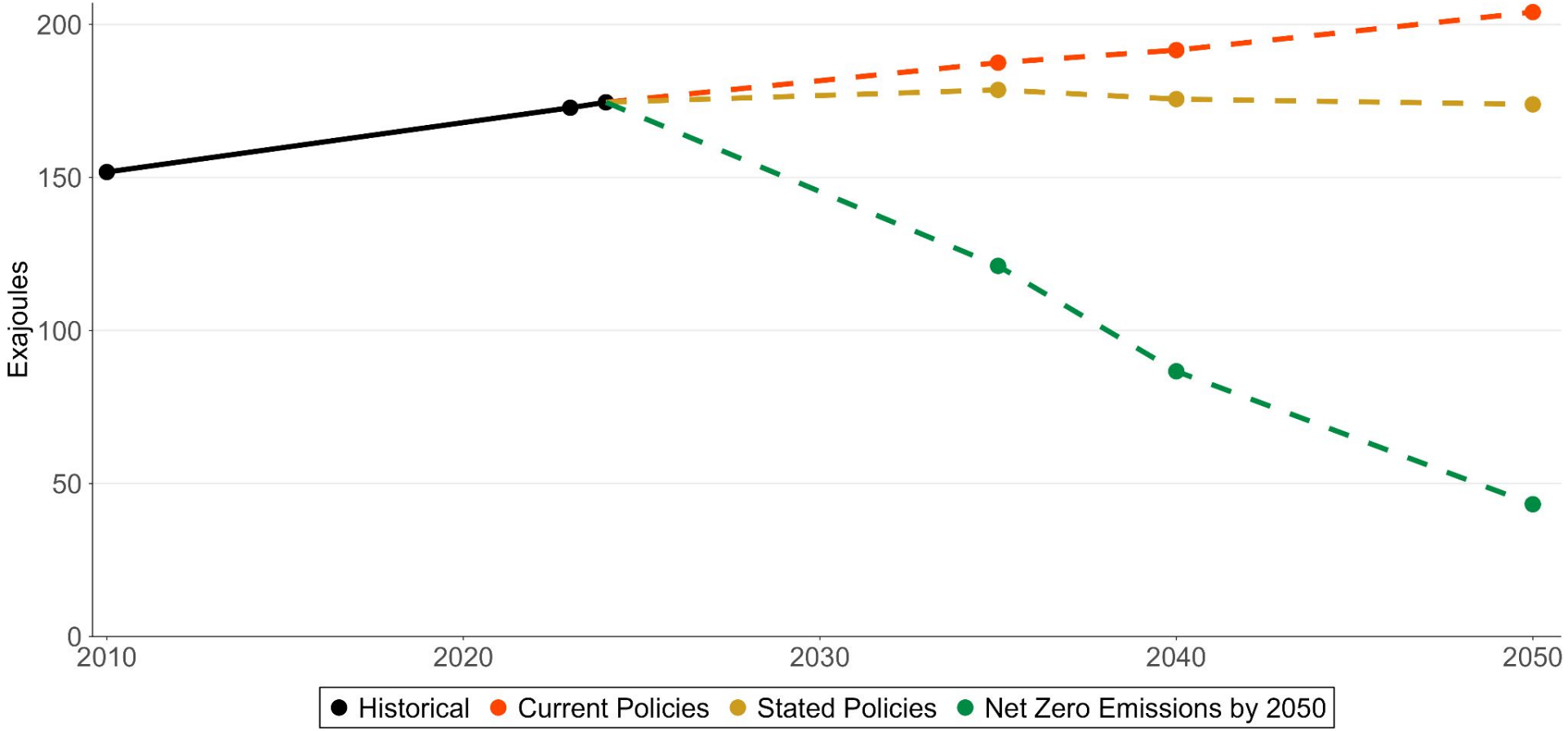
1963 to 2024



Source: Organization of the Petroleum Exporting Countries

Energy Consumption from Oil

By Scenario



Source: International Energy Agency

Why is this important?

→ Promotes stability (and inertia) in the energy sector

→ Underscores importance of policy for decarbonization

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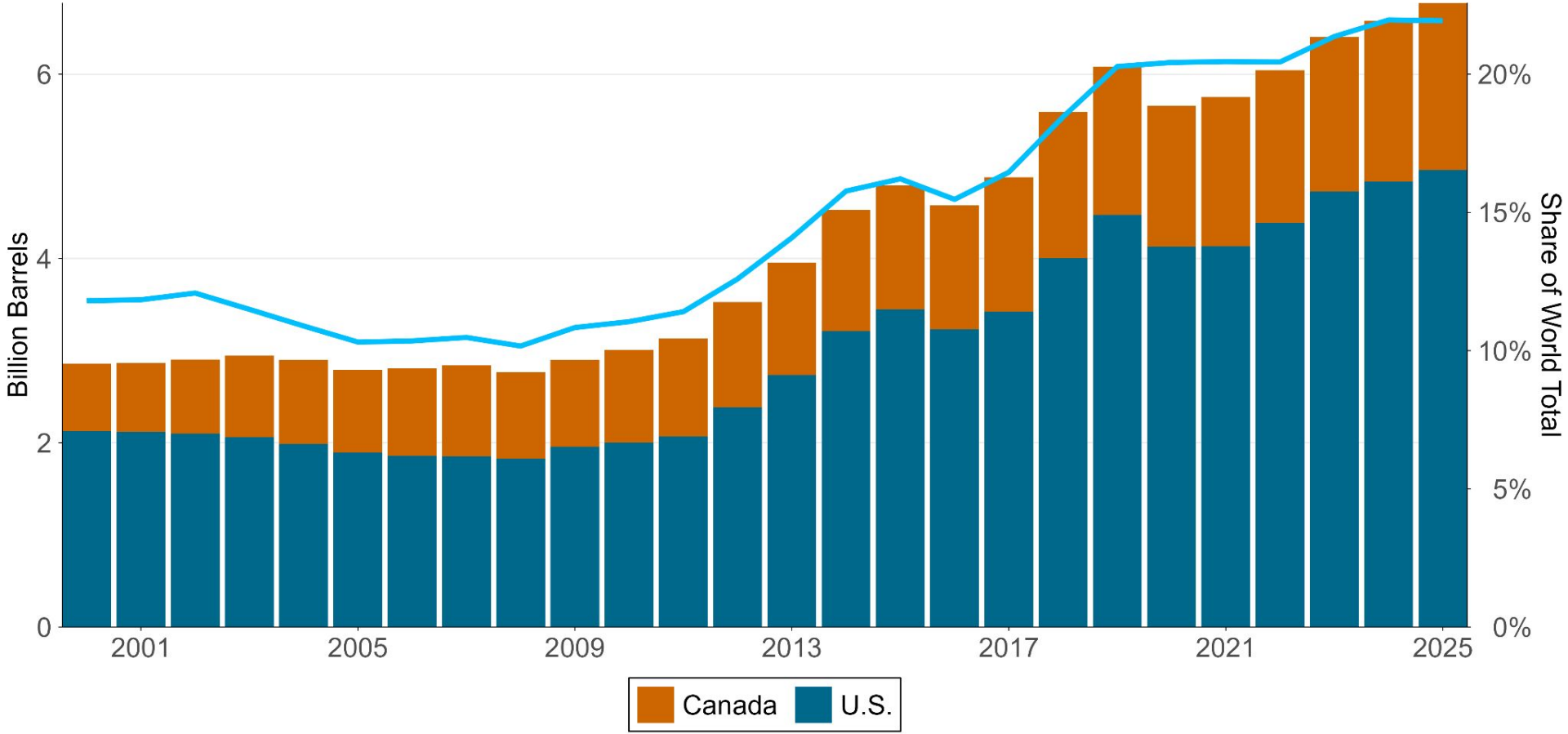
Oil Production in Canada and the U.S.

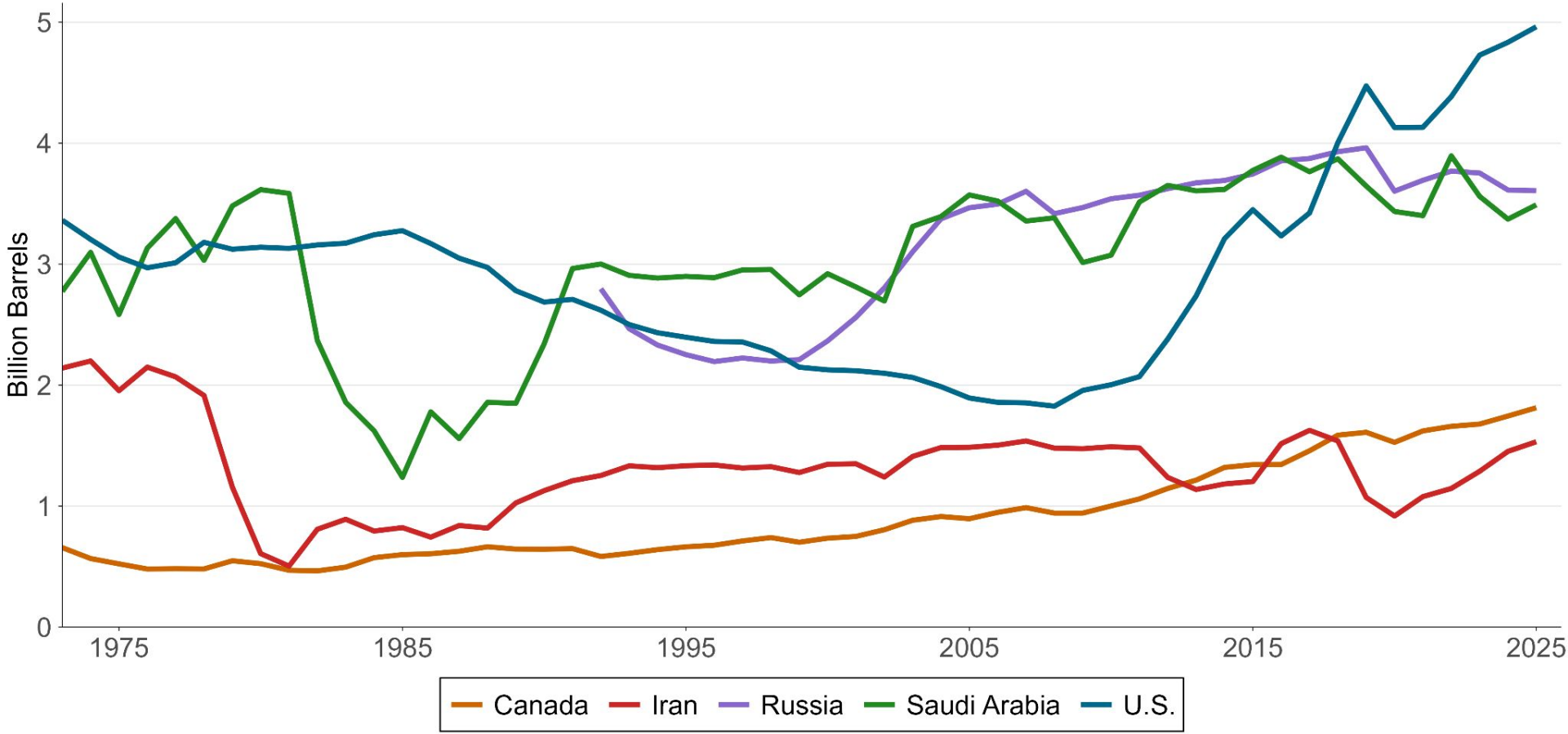
2000-Present



Source: Energy Information Administration

Oil Production in Selected Countries

1973-Present



Source: Energy Information Administration

Why is this important?

→ Mutes impact of oil supply shocks

→ Increases American geopolitical leverage

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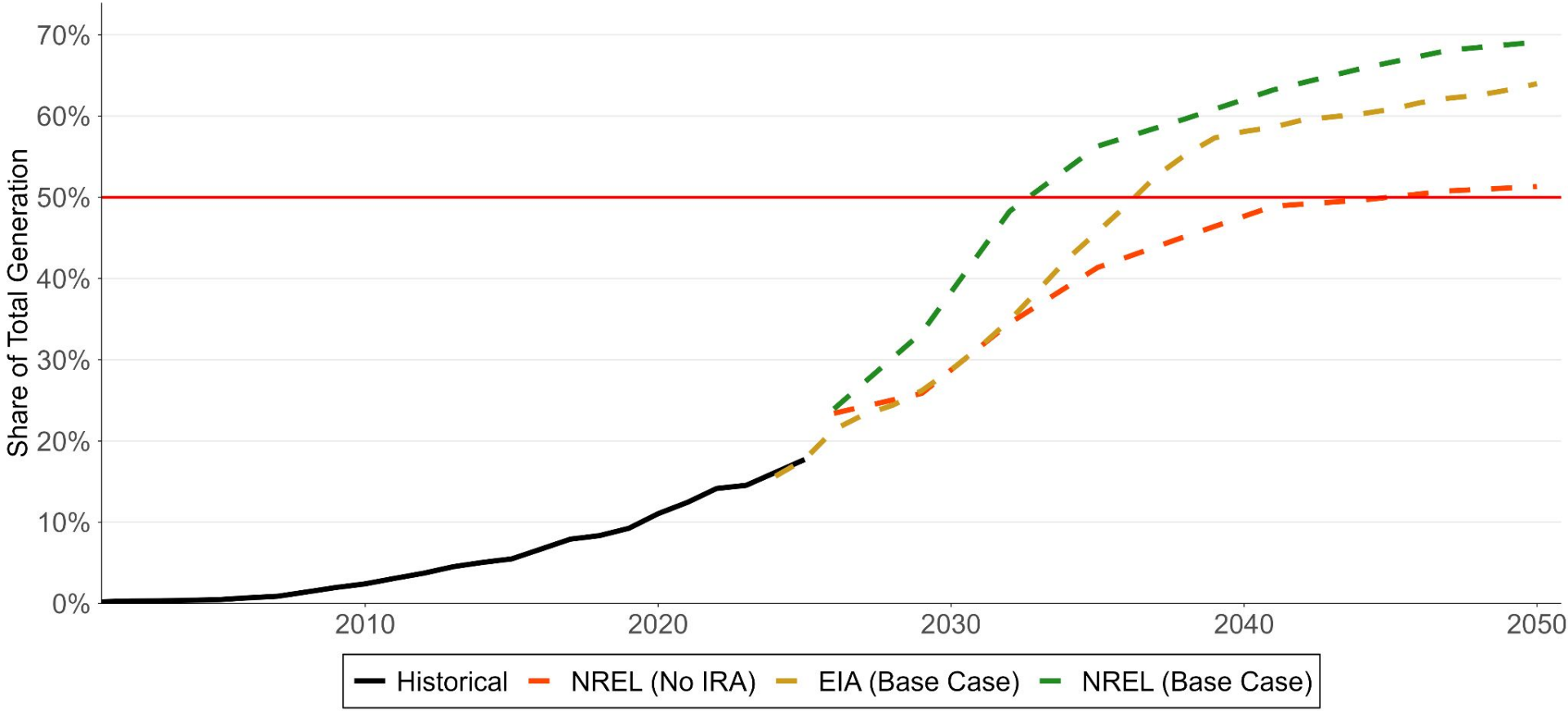
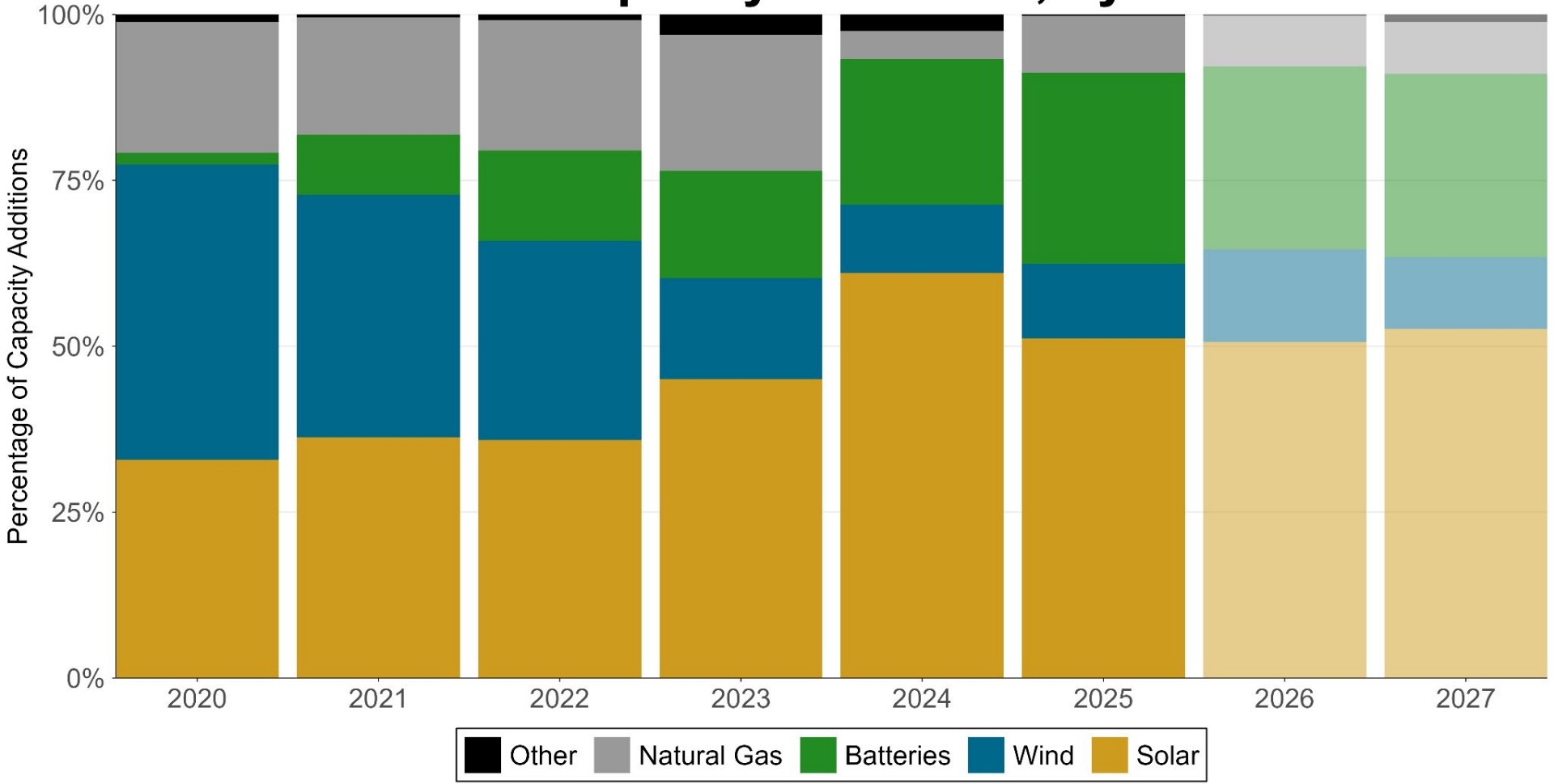
Electricity Generation from Wind and Solar

By Scenario



Sources: National Renewable Energy Laboratory and Energy Information Administration.
Note: Utility-scale, power-sector generation only.

Electric Capacity Additions, by Source



Source: Energy Information Administration.

Levelized Cost of Energy Comparison—Version 18.0

Selected renewable energy generation technologies remain cost-competitive with conventional generation technologies under certain circumstances

Source: Lazard estimates and publicly available information.

Why is this important?

→ Requires large-scale buildout of generation and transmission infrastructure

→ Increases feasibility of electrification as a decarbonization pathway

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